

### 6.2.2 Coulomb's law

| Learning outcomes                                                                                                          | Additional guidance                              |
|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                              |                                                  |
| (a) Coulomb's law; $F = \frac{Qq}{4\pi\epsilon_0 r^2}$<br>for the force between two point charges                          | <b>THIS WILL ALSO REQUIRE KNOWLEDGE OF 3.2</b>   |
| (b) electric field strength $E = \frac{Q}{4\pi\epsilon_0 r^2}$<br>for a point charge                                       |                                                  |
| (c) similarities and differences between the gravitational field of a point mass and the electric field of a point charge. | <b>THIS WILL ALSO REQUIRE KNOWLEDGE OF 5.4.2</b> |